Prioritization of Quality Requirements in Decentralized Identity Applications: What Matters Most to Issuers?

Welcome, and thank you for taking part in this survey, which is a central part of a Master Thesis project conducted at the Communication Systems Group (CSG) within the Department of Informatics at the University of Zurich.

What is this survey about?

This Master Thesis aims to prioritize the “quality requirements” for modern digital identity systems, specifically those based on Decentralized Identity (DI) and Self-Sovereign Identity (SSI). In software development, we often distinguish between “what” a system does (Functional Requirements) and how it accomplishes these tasks. These “how” aspects — usually referred to as Non-Functional Requirements (NFRs) or quality requirements — describe abstract yet critical qualities, such as such as consent, security, authenticity, and privacy.

Specifically, it focuses on systems built on two key concepts: Self-Sovereign Identity (SSI) and Decentralized Identity (DI).

What is Self-Sovereign Identity (SSI)?

Imagine your users holding a digital wallet that contains their various forms of identification (like driver’s licenses, university degrees, or employee badges) securely on their device. SSI is a model designed to give holders complete control over that digital identity. The core goals are to enhance holders’ control over their data, guarantee its privacy, and improve trust in digital interactions. Instead of relying on a single central provider, holders decide which information to share and with whom.

What is Decentralized Identity (DI)?

DI is the underlying technology that enables SSI. Traditionally, digital identities are stored in centralized systems managed by companies or governments. A decentralized approach means there is no single point of control or failure. This architecture involves three key roles:

- **Issuers** who create and issue digital credentials
- **Holders** who store and present them
- **Verifiers** who check them

Emerging SSI/DI systems aim to give holders more control over their data, enhance privacy, and foster trust in digital interactions. This research focuses on translating these abstract goals into practical terms and understanding which qualities matter most to each role.

To do this, we are gathering feedback from the three leading roles within a decentralized identity ecosystem:

- **Holders**: Individuals who own and control their identity data.
- **Issuers**: Organizations (like universities, governments, or businesses) that provide credentials.
- **Verifiers**: Organizations (like employers or service providers) that request and verify credentials.

What will you be asked to do?

This questionnaire is designed to understand **issuers'** priorities regarding the quality requirements of Decentralized Identity systems and should take **about 10 minutes** to complete.

The survey consists of two main parts:

1. You will be asked to rate the importance of several quality requirements (e.g., privacy, security).
2. You will be asked to choose the most and least important qualities from a given list.

There are no right or wrong answers. We are interested in your professional judgment as an **issuer**. Your participation is highly valued and will contribute significantly to this research.
There are 40 questions in this survey.

Participant background information

To help us interpret the survey results correctly, we first ask a few brief questions about your background – such as gender, year of birth, professional role, and any prior experience with Self-Sovereign Identity (SSI) or Decentralized Identity (DI).

Privacy & Confidentiality

This survey is conducted as part of a Master Thesis at the University of Zurich, and all responses are recorded fully anonymously.

- No personal identifiers: Neither names nor contact details are requested.
- **Anonymous analysis:** All data are stored and reported in aggregate form, preventing any individual identification.
- **Legal basis:** Data collection complies with the cantonal Data Protection Act (IDG) and the UZH Policy on Ethical Review of Research Projects Involving Human Subjects.
- **Voluntary participation:** You may withdraw or discard your responses at any time before submitting the survey.

Answering these items is voluntary, anonymous, and will be used **only** to analyse how perceptions of non-functional requirements may vary across different participant groups.

Please indicate your gender.

Please choose **only one** of the following:

- ◯ Female
- ◯ Male

Please enter your year of birth (YYYY).

Please write your answer here:

Which of the following best describes your current occupation or position? (Select all that apply.) *

Please choose **all** that apply:

- [ ] In training (Student / Apprentice)
- [ ] Manager / Executive
- [ ] Professional / Academic occupation
- [ ] Technician and equivalent occupations
- [ ] Clerical and administrative occupations
- [ ] Service and sales worker
- [ ] Skilled worker in agriculture and forestry
- [ ] Craft and related trades
- [ ] Plant and machine operator / Assembler
- [ ] General laborer

If you are representing a company or institution in your role as issuer, please feel free to specify its name below. Otherwise, you may leave this blank to remain anonymous.

Please write your answer here:

If you wish, please describe your professional role or position (e.g., "Credential Services Manager"). You may leave this blank if you prefer not to say.

Please write your answer here:

Background

Self-Sovereign Identity, or SSI, is a new approach that puts **you** in control of your own digital identity. SSI is an approach to digital identity in which individuals (or organisations) create, store and present their own verifiable credentials without relying on a single central provider. The main idea is that your identity should belong to you, just like your physical passport or driver's license. Instead of logging in through a company, you would manage your identity from a secure personal app on your phone, often called a digital wallet.

A Decentralized Identity (DI) is your digital identity that you own and control entirely. Unlike an email address or username, which are issued and managed by specific companies, a DI is not tied to any single provider and cannot be revoked by anyone else. With your DI, you can prove who you are and share verified information from your digital wallet in a way that is secure, private, and entirely under your authority, giving you actual ownership of your digital self.

Have you ever used, implemented, or otherwise worked with SSI or DI?

*

Please choose **only one** of the following:

- ◯ Yes
- ◯ No

Question Group A – Key Qualities of Credential Issuance

On the next pages you will see short statements about **important qualities in issuing digital credentials** (for example, “Credentials can always be verified by holders”). For **each statement**, please tell us how strongly you agree or disagree on this 1-to-5 scale:

- 1 = Strongly disagree 5 = Strongly agree

we are interested in your professional perspective as an **issuer** on how important these core qualities are.

The ability to digitally sign each credential I issue to guarantee its authenticity and integrity is important to me.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

Issuing credentials without a verifiable digital signature would be acceptable to me.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I need to issue credentials that work across both modern identity platforms and traditional identity systems.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

Credentials that are incompatible with existing legacy systems would be acceptable.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I need to obtain explicit permission from the holder before issuing any credential about them.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

Issuing a credential without the holder's clear consent would be tolerable.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	◯	◯	◯	◯	◯	**Strongly agree**

I want the credentials I issue to work with any system or service, no matter the platform.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

It would be manageable if some services rejected the credentials I issue.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I need to define and publish my credential schemas in widely accepted formats so that any system can process the credentials I issue.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I could tolerate credentials with non-standard or outdated schemas being unusable by many services.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

Question Group B – Additional Qualities of Credential Issuance

You will now see another set of statements describing **other possible qualities** of issuing digital credentials (for example, ease of revoking or updating credentials).
Please follow the same procedure:

1. Read each statement.
2. Use the 1-to-5 scale to show how strongly you agree or disagree.

Answer every item according to your professional experience or expectations as an **issuer**. Your responses will help us understand which additional qualities issuers value most in future digital-identity systems.

I need to include only the personal data strictly necessary when issuing a credential.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

Including any unnecessary personal data in a credential would be acceptable.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I need identity data to be safeguarded from misuse during the credential-issuance process.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

The possibility of credentials I issue being intercepted or tampered with would be an acceptable risk.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I need to record each credential I issue on a public, tamper-proof ledger so anyone can audit my actions.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

A lack of a tamper-proof record for the credentials I issue would be an acceptable situation.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I must digitally sign each credential and publish its public key or status so it can be independently verified.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I would remain confident in the credentials I issue even if they lack a verifiable signature or current status.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I need to issue credentials without incurring high transaction or storage fees.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

A system that charges high fees for issuing credentials would be a minor factor in my decision to use it.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I need to issue and revoke credentials using a decentralized registry without relying on any single central authority.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I feel comfortable issuing credentials that rely on a single central provider.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

I must ensure that identity data are secure against threats during credential issuance.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

Security measures during credential issuance feel nonessential to me.

*

Please choose the appropriate response for each item:

	1	2	3	4	5	
Strongly disagree	○	○	○	○	○	**Strongly agree**

Priorities: What matters most — what matters least

Prioritisation of Non-Functional Requirements (NFR) in Self-Sovereign Identity (SSI) and Decentraized Identity (DI)

Over the next few pages you will complete **9 short Best–Worst choice tasks**.
Each task presents **four qualities (non-functional requirements, NFRs)** that can shape the **credential-issuance process** in a decentralized-identity ecosystem—for example, Security, Cost, Privacy or Interoperability.

For every set of four:

- **Select the single quality that, in your view, is *most important*.**

- **Select a different quality that is *least important*.**

You must choose one “Most important” and one “Least important” on every screen; the same item cannot be both. There are no right or wrong answers, and the full section takes about 2 to 3 minutes.

Description

Authenticity: *Digital signature guarantees credential origin and integrity.*

Compatibility: *Credentials work with both SSI wallets and legacy PKI systems.*

Consent: *Explicit GDPR-compliant consent is obtained before issuing credentials.*

Interoperability: *Credentials are usable across all platforms and services.*

*

Please choose the appropriate response for each item:

	Most important	Least important
Authenticity	○	○
Compatibility	○	○
Consent	○	○
Interoperability	○	○

Description

Privacy: *Only the data strictly necessary for issuance is stored and processed.*

Protection: *Credentials are protected from misuse during issuance.*

Security: *Holder is authenticated and data secured before issuing.*

Standard: *Credentials and schemas comply with recognized open standards*

*

Please choose the appropriate response for each item:

	Most important	Least important
Privacy	○	○
Protection	○	○
Security	○	○
Standard	○	○

Description

Transparency: *Issuance processes are transparent and auditable.*

Verifiability: *Credentials remain verifiable via signatures and revocation lists.*

Cost: *Issuance and integration costs are minimized.*

Decentralization: *Decentralized registries are used; central control is avoided.*

*

Please choose the appropriate response for each item:

	Most important	Least important
Transparency	○	○
Verifiability	○	○
Cost	○	○
Decentralization	○	○

Description

Authenticity: *Digital signature guarantees credential origin and integrity.*

Privacy: *Only the data strictly necessary for issuance is stored and processed.*

Cost: *Issuance and integration costs are minimized.*

Interoperability: *Credentials are usable across all platforms and services.*

*

Please choose the appropriate response for each item:

	Most important	Least important
Authenticity	○	○
Privacy	○	○
Cost	○	○
Interoperability	○	○

Description

Compatibility: *Credentials work with both SSI wallets and legacy PKI systems.*

Protection: *Credentials are protected from misuse during issuance.*

Transparency: *Issuance processes are transparent and auditable.*

Security: *Holder is authenticated and data secured before issuing.*

*

Please choose the appropriate response for each item:

	Most important	Least important
Compatibility	○	○
Protection	○	○
Transparency	○	○
Security	○	○

Description

Consent: *Explicit GDPR-compliant consent is obtained before issuing credentials.*

Standard: *Credentials and schemas comply with recognized open standards.*

Verifiability: *Credentials remain verifiable via signatures and revocation lists.*

Decentralization: *Decentralized registries are used; central control is avoided.*

*

Please choose the appropriate response for each item:

	Most important	Least important
Consent	○	○
Standard	○	○
Verifiability	○	○
Decentralization	○	○

Description

Authenticity: *Digital signature guarantees credential origin and integrity.*

Protection: *Credentials are protected from misuse during issuance.*

Decentralization: *Decentralized registries are used; central control is avoided.*

Verifiability: *Credentials remain verifiable via signatures and revocation lists.*

*

Please choose the appropriate response for each item:

	Most important	Least important
Authenticity	○	○
Protection	○	○
Decentralization	○	○
Verifiability	○	○

Description

Compatibility: *Credentials work with both SSI wallets and legacy PKI systems.*

Standard: *Credentials and schemas comply with recognized open standards.*

Interoperability: *Credentials are usable across all platforms and services.*

Privacy: *Only the data strictly necessary for issuance is stored and processed.*

*

Please choose the appropriate response for each item:

	Most important	Least important
Compatibility	○	○
Standard	○	○
Interoperability	○	○
Privacy	○	○

Description

Consent: *Explicit GDPR-compliant consent is obtained before issuing credentials.*

Cost: *Issuance and integration costs are minimized.*

Security: *Holder is authenticated and data secured before issuing.*

Transparency: *Issuance processes are transparent and auditable.*

*

Please choose the appropriate response for each item:

	Most important	Least important
Consent	○	○
Cost	○	○
Security	○	○
Transparency	○	○

You have successfully completed the survey.

Thank you very much for your valuable time and participation. With your detailed answers, you have made a crucial contribution to my Master's thesis, which focuses on the prioritization of software quality requirements for future digital identity systems (Self-Sovereign Identity). Your perspective is essential for this research.

If you have any further questions, comments or remarks about this survey, please feel free to enter them in the text field below.

Please write your answer here:

Thank you for your time and valuable contribution to this research.

The data you have provided will be analyzed to understand how different groups prioritize qualities in Decentralized Identity systems.

Your input is essential for fulfilling the goals of this Master Thesis. We appreciate your willingness to share your perspective.

Submit your survey.
Thank you for completing this survey.